

Dear Algo: A Precision-First Agentic Intent Layer for Unified Search and Recommendation

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Abstract

Search and recommendation serve a shared discovery objective but encode intent differently. We study this boundary through Dear Algo on Threads, a deployed product where open-ended requests such as *more NBA news* or *less politics* steer subsequent feed recommendations rather than return a one-shot result list. Its agentic intent layer compiles explicit, inferred, negative, and compound intent into a grounded executable plan, then invokes conventional retrieval and optional semantic or multimodal reranking. The layer shares an intent-to-retrieval contract without requiring one model or serving path across search-like and recommendation-like modes.

We evaluate Dear Algo under a precision-first objective. In a blinded audit of 300 public request-item pairs (296 evaluable), a strict categorical LLM-as-a-judge gate achieved 94.4% exact-Relevant precision [88.8%, 98.9%]. Across 72 normalized request clusters, the full configuration produced 7.73 judge-qualified candidates per 20 slots versus 6.61 for an LLM-derived-query baseline, a gain of 1.11 [0.12, 2.12]. In a candidate-randomized serving-path study restricted to the reranker path’s first 72 eligible hours, the user-weighted judge-Irrelevant share among judged admissions was 2.80% versus 4.78% off (-1.97 points [-3.02, -0.94]), while Exact-Relevant share was 2.24 points higher [0.08, 4.41].

Together, these studies show how explicit natural-language intent can be carried into feed recommendation under a precision-first evaluation framework.

CCS Concepts

• **Information systems** → **Recommender systems; Retrieval models and ranking; Evaluation of retrieval results**; • **Computing methodologies** → *Natural language processing*.

Keywords

unified search and recommendation, agentic retrieval, intent understanding, retrieval-augmented generation, structured retrieval, precision-first evaluation, industrial recommender systems

1 Introduction

Search exposes explicit but typically short-lived intent through a query and is commonly evaluated by query-item relevance. Recommendation builds longer-lived preferences primarily from implicit actions such as views, shares, skips, and replies and is often evaluated through engagement. Even over a shared catalog, an explicit request is rarely converted into durable, executable state for feed

recommendation. LLMs enable open-ended interpretation, but production retrieval still requires valid identifiers, explicit constraints, bounded execution, fallback, and inspectable telemetry.

Dear Algo is a deployed Threads interface at this boundary. A user can ask for *more NBA news*, *less politics*, or content that will *make me laugh*; the instruction steers subsequent feed retrieval rather than returning only a one-shot ranked list [32, 53, 54]. Figure 1 shows the public request and settings surfaces.

Dear Algo uses an agentic intent layer built on the SilverTorch serving substrate [66]. The layer interprets an instruction into typed positive and negative constraints, grounds open-ended phrases to a controlled item-side semantic vocabulary, and compiles a reusable SilverTorch Query Language (STQL) plan for candidate retrieval and Boolean filtering. STQL is an internal JSON-serialized retrieval DSL, not relational SQL. Optional semantic or multimodal reranking operates on a bounded candidate set.

The compiler and plan are shared, while interfaces, operating points, and serving paths may remain mode-specific. This modularity matters because the product computational cost is asymmetric: an irrelevant item admitted into a persistent feedback loop can cause repeated unwanted exposure, whereas rejection mainly reduces coverage. We therefore separate admission precision, recall, ranking and engagement, and then report user actions accordingly.

This paper contributes:

- (1) **A shared executable intent layer for feed steering.** Dear Algo maps explicit requests and inferred preferences into one grounded plan while retaining mode-specific serving paths.
- (2) **A layered account of unification.** We distinguish infrastructure, representation, and empirical unification and identify which layers the system is designed and implemented to share.
- (3) **A precision-first evaluation bridge.** We combine blinded human calibration, offline candidate evaluation, randomized admitted-set analysis, and behavioral analysis without treating their estimands as interchangeable.
- (4) **Industrial evidence and measurement lessons.** We report human calibration, paired offline retrieval, candidate-randomized admitted-set quality, and observational usage, and identify the logging required for causal cross-mode evaluation.

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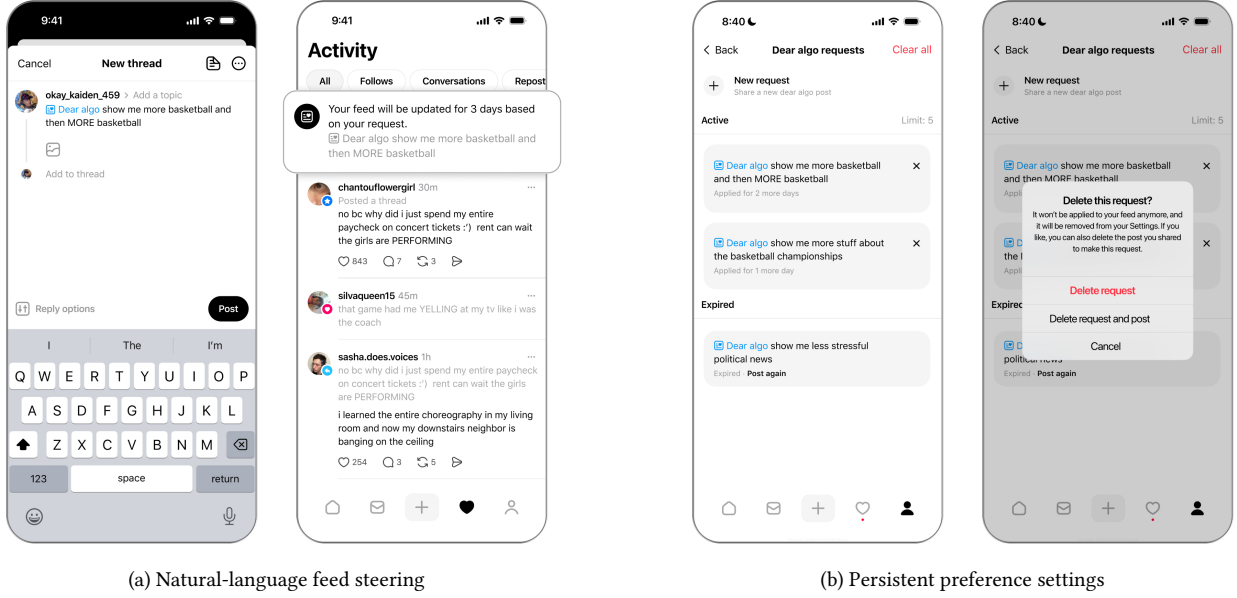


Figure 1: Dear Algo product surfaces on Threads. Users express feed preferences through natural-language requests and review or manage the resulting controls in Settings. Images reproduced from the public Meta Newsroom announcement [32].

2 Related Work

2.1 Unified Search and Recommendation

Search and recommendation address a common discovery problem over shared catalogs, but condition their decisions on different evidence: search starts from an explicit, often transient query, whereas recommendation infers preferences from interaction history. Existing work unifies behavior sequences, graphs, encoders, or transition models [45, 65, 67, 73], and KuaiSAR provides aligned logs for studying both tasks [50]. Other approaches share item representations [69, 70], transfer search representations into recommendation [47, 72], or use generative approaches to jointly model both tasks [29, 37, 46].

Cross-task transfer is not automatically beneficial. Joint gains depend on aligned histories, compatible popularity distributions, and related item co-occurrence structure; semantic identifiers optimized for one task can degrade the other [37, 38, 46]. Our system therefore shares the intent-to-serving contract—a controlled vocabulary and typed executable plan—while retaining mode-specific retrieval, ranking, thresholds, and latency budgets. Whether a plan originating in one mode causally improves another remains a separate empirical question.

2.2 Interactive and Steerable Recommendation

Interactive recommendation predates LLMs. Critiquing systems let users refine attributes, and conversational systems interleave preference elicitation with recommendation [9, 20]. Scrutable profiles expose preferences for inspection and editing [4, 34, 40], while instruction-following models provide language-based control over recommendations [31, 71]. CTRL-Rec, RecBot, and SteerEval more directly study natural-language steering [7, 51, 76].

Dear Algo’s distinction is operational: its intent layer grounds positive, negative, and compound instructions to production feature identifiers and compiles them into an inspectable serving plan. Because that plan can outlive the initiating request, its lifecycle must handle preference drift and responsiveness to correction [25, 44, 63]. Provenance and deletion are additional state-management requirements.

2.3 Grounded Agentic Retrieval

The intent layer is related to semantic parsing, entity linking, and constrained decoding, which separate schema or identifier validity from semantic correctness [10, 41, 58, 64]. Query-expansion methods instead generate free-form retrieval text [16, 59], while generative retrieval produces item identifiers directly [39, 52]. The intent layer keeps the catalog external: an LLM proposes intent, grounding resolves supported values, and a typed DSL expresses predicates and retrieval parameters.

Agentic recommenders commonly place an LLM control plane over conventional retrieval and ranking tools [19, 62, 74]. Dear Algo follows this pattern rather than replacing the serving data plane; structured predicates execute over a neural candidate service, analogous to filtered vector search [17, 36]. Its bounded plan-and-execute path is narrower than general tool-using and adaptive reasoning-and-retrieval loops [3, 56, 68]. Classical RAG conditions a generated answer on passages [27]; the grounding stage instead retrieves vocabulary entries before item retrieval. Conversational recommendation and LLM reranking operate at still different boundaries [49, 61].

2.4 Evaluation Across Paradigms

Information retrieval (IR) test collections estimate ranking effectiveness for fixed topics and a fixed corpus; incomplete pools and assessor disagreement bound that estimand [6, 57]. Offline recommender benchmarks instead predict held-out behavior under a historical exposure and candidate protocol, where unexposed items are not valid negatives and sampled metrics may change conclusions [21, 26, 42, 43]. Neither design by itself estimates the value of deploying a new cross-mode policy.

Counterfactual estimators require logged propensities and support for the target actions [13, 28]; those requirements are not met by the observational study reported here. Randomization most directly identifies a deployable policy effect [24], but sessions or clicks still need not measure intent fulfillment, and short experiments can miss recommendation feedback loops and long-run welfare [8, 33]. We therefore keep request relevance, admitted-set quality, user behavior, and report actions as separate estimands.

2.5 LLM-as-a-Judge for Relevance Evaluation

LLM judges are measurement instruments, not task-independent oracles. Pointwise labels, pairwise preferences, and scalar scores answer different questions [23, 30, 75]; their behavior varies with the model, prompt, rubric, scale, decoding, threshold, and population [1, 2, 14, 55]. Documented presentation-order, length, and self-preference effects further motivate in-domain validation rather than wholesale human replacement [12, 35, 48, 60].

Validation therefore attaches to the complete instrument $J = (\text{model/version, prompt, rubric, scale, decoding, threshold})$. Agreement or system-level correlation does not calibrate a numeric score or a new operating point [5, 11, 15, 18]. For a selective gate, admission precision, coverage, and relevant recall are distinct; prospective guarantees additionally require a frozen threshold-selection procedure [22].

Accordingly, our blinded audit validates the production LLM judge scorer at the strict categorical operating point on the sampled public population. The paired Top-20 study uses the same scorer but reports fixed-denominator *judge-qualified yield*, not human precision. The candidate-randomized study also uses this scorer, but estimates quality among judged admissions within the serving window rather than quality over an unobserved pre-reranking slate.

3 Dear Algo: A Layered Intent Architecture

Figure 2 summarizes the system’s compilation and execution path.

3.1 Levels of Unification

We distinguish three claims:

- (1) **Infrastructure unification:** modes share orchestration, catalogs, or serving components.
- (2) **Representation unification:** modes compile intent into the same semantic vocabulary and executable plan.
- (3) **Empirical unification:** information originating in one mode causally improves another mode without unacceptable regression.

Dear Algo is designed and implemented to share the first two layers. Catalog and vocabulary identifiers, intent interpretation,

and STQL plans are common; candidate configuration, ranking, and interaction surfaces remain mode-specific. The experiments in Sections 4 and 5 do not establish causal empirical unification.

3.2 Intent Compilation

Let q be an optional explicit request, h_u eligible user context, c the immediate interaction context, and m the discovery mode. The intent layer receives

$$x = (q, h_u, c, m), \quad m \in \{\text{explicit, passive, negative, compound}\}. \quad (1)$$

It produces a normalized intent z , executable plan p , candidates C , and ranking R :

$$\begin{aligned} z &= f_{\text{intent}}(x), \\ p &= f_{\text{ground}}(z, V), \\ C &= f_{\text{retrieve}}(p), \\ R &= f_{\text{rank}}(C, z, h_u), \end{aligned} \quad (2)$$

where V is the controlled semantic vocabulary.

The interpreter emits schema-constrained semantic features rather than an unconstrained retrieval string. Each feature has an explicit categorized semantic dimension (e.g., topic, language, location, or freshness), polarity where applicable, candidate values, and type-specific valid operators. Grounding retrieves supported vocabulary values and identifiers. The planner then composes inclusion, exclusion, conjunction, disjunction, language, location, and freshness constraints into STQL.

Running example. Consider the synthetic request *Dear Algo, show me more women’s basketball analysis, but fewer score spoilers*. The interpreter emits two positive features and one negative feature. Grounding maps the three phrases to controlled identifiers v_1 , v_2 , and v_3 . A human-readable view of the resulting plan is

$$p_{\text{ex}} = \text{AND}(\text{TAG}(v_1), \text{TAG}(v_2), \text{NOT}(\text{TAG}(v_3))) . \quad (3)$$

Production serialization replaces readable labels with controlled identifiers and encodes the Boolean tree for model-query execution. Execution accepts only typed plans whose operators and grounded values satisfy the serving contract; unsupported plans abstain or follow the existing fallback path.

3.3 Execution and Persistence

The STQL plan is sent to an existing high-throughput candidate generation service. Candidates may be scored against the normalized intent and optionally reranked by a text or multimodal LLM. Expensive stages are restricted to a bounded slate and can be skipped on passive or latency-sensitive paths. The plan can be persisted and refreshed against new content without rerunning every reasoning stage. Architecturally, this separates intent compilation as a control-plane concern from candidate retrieval as a latency-sensitive data-plane concern. The design follows a compile-once, execute-many principle: expensive intent reasoning is amortized across refreshes, while execution remains late-bound to current inventory. Optional bounded reranking permits latency-sensitive paths to skip expensive stages without changing the shared plan contract.

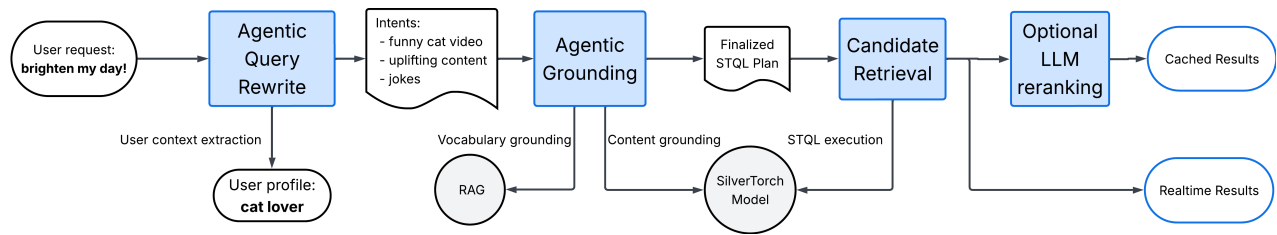


Figure 2: Dear Algo’s intent layer compiles heterogeneous intent into a shared grounded STQL plan that drives candidate retrieval, scoring, reranking, and plan lifecycle operations.

3.4 Implementation Substrate

Dear Algo’s intent-layer implementation is called SilverTorch Agentic Recommender (STAR). It runs on SilverTorch, a model-based GPU recommendation serving system that integrates filtering, approximate nearest-neighbor retrieval, and scoring as model components [66]. An explicit request can be executed immediately and optionally persisted; for feed recommendation, the persisted or inferred plan can be refreshed against new inventory to produce intent-conditioned candidates for the existing ranking pipeline. Negative intent becomes suppression clauses, while compound intent is decomposed into unit plans whose results are deduplicated and aggregated. Dear Algo therefore connects search-like requests with recommendation-like continuous retrieval without replacing the downstream recommender or requiring both modes to share one ranking model. The explicit surface is not general web search.

4 Evaluation

4.1 Research Questions and Evidence Tiers

We ask:

- **RQ1:** Can the evaluator support a high-precision categorical gate for request-item relevance?
- **RQ2:** Holding the LLM-derived query fixed, does the full grounded-tag policy produce more strictly judge-qualified candidate yield than the retrieved-tag policy?
- **RQ3:** Within the delivery-eligible window, does randomized LLM reranking improve precision-first quality among judged admitted items?
- **RQ4:** Is explicit-request initiation followed by higher later Threads app usage, and how does the report rate change?

The studies answer different questions. Human calibration validates one measurement operating point. The offline study evaluates paired candidates from a frozen request-time pool. The candidate-randomized study estimates judge-labeled quality among admitted impressions within serving common support. The behavioral study measures within-user change around an explicit request but lacks a treatment control. We do not combine them into a single significance claim.

4.2 Blinded Human Calibration

We formed a frozen sampling frame of 48,876 non-employee public-mode request-item evaluation records logged from July 17 through July 24, 2026, produced by the locked categorical judge configuration and excluding records used in prior pilot samples. The frame

contained 2,212 judge-Irrelevant, 14,138 judge-Partial, and 32,526 judge-Relevant records. We deliberately sampled 90, 120, and 90 pairs from these strata, respectively. This design oversamples the rarer Irrelevant and boundary Partial strata for audit precision; cross-stratum population estimates therefore use inverse selection weights. Six blinded annotation workbooks yielded 900 initial ratings. Of 300 pairs, 118 had unanimous labels. A seventh independent blinded rating supplied the final label for 178 evidence-ready disagreements. Four pairs with insufficient evidence (e.g. due to privacy constraints) were excluded and reported, leaving 296 pairs from 296 users.

The audited instrument is the locked, pointwise production LLM judge scorer. For the alignment audit, its logged outputs are represented by the frozen Irrelevant, Partial, and Relevant strata. The strict gate admits only Relevant; its primary estimand is exact-Relevant admission precision:

$$P(\text{human Relevant} \mid \text{judge Relevant}). \quad (4)$$

The production prompt, model version, decoding configuration, and service details are proprietary and are not disclosed. At-least-Partial precision is a severe-false-admission sensitivity. Exact Relevant recall is reported as a diagnostic, not as the product objective. Strict-gate precision conditions entirely on the judge-Relevant stratum, so its inverse-probability weight cancels. Metrics that combine judge strata use the frozen inverse selection probabilities to recover the eligible target frame. Intervals use 10,000 sampling-stratum-preserving user-cluster bootstrap replicates with those weights. The zero severe-false-admission count also receives an exact binomial interval. This primary analysis uses the frozen adjudicated labels and applies no latent-label or rater-bias adjustment.

Separately, we fit a post-hoc smoothed Dawid–Skene model by expectation-maximization to examine rater-severity sensitivity. The model uses the balanced assignments across the six original blinded raters to estimate per-item latent-class probabilities and treats insufficient evidence as an abstention response. The selectively assigned seventh rater is excluded. Each of 10,000 sampling-stratum-preserving bootstrap replicates resamples pairs and refits the model. This exploratory, model-based sensitivity does not alter the frozen labels or primary estimate and does not provide a prospectively calibrated pass threshold.

Table 1: Paired offline retrieval configuration arms.

Arm	Query text	Grounded-tag policy
A	Raw request	Use retrieved canonical tags
B	LLM-derived query	Use retrieved canonical tags
C	LLM-derived query	Full LLM-selected grounded-tag policy

Note: All displayed arms use the same substrate and request-time inventory.

4.3 Paired Offline Candidate Study

The offline dataset contains 100 request events from 98 users. Ninety-two produced valid continuous judge responses and were aggregated into 72 normalized request-text clusters. Eight malformed JSON responses were excluded symmetrically from the primary analysis; a sensitivity assigns zero yield to every arm for those events. Every arm executes on the same SilverTorch candidate-serving substrate, uses the same request-time available item pool, and has a fixed Top-20 budget. Infrastructure, inventory, and the slot budget are held fixed; each configuration may retrieve a different slate. Table 1 defines the displayed configurations.

All displayed configurations use tag grounding and compile the result through STQL. C-B is therefore a configuration-level contrast between two grounded-tag policies after holding the LLM-derived query fixed; it does not isolate any individual grounding mechanism.

For request i and arm a , the fixed-denominator outcome is

$$Y_{i,a}@20 = \frac{1}{20} \sum_{j=1}^{20} \mathbf{1}[s_{i,a,j} \geq 0.8]. \quad (5)$$

Let $Q_{i,a}$ denote the numerator above. Table 3 reports the primary outcome as both Yield@20, $Y_{i,a}@20$, and qualified candidates@20, $Q_{i,a}$; these are the same fixed-denominator outcome in rate and count units, respectively. Here $s_{i,a,j}$ is the continuous judge score on $[0, 1]$. The value 0.8 was a pre-existing internal product-review quality bar used in the initial offline evaluation, before C-B was selected as the focal paper contrast. We retained it post hoc as the qualification gate for the precision-first analysis. The cutoff was not tuned to the C-B contrast, selected from human labels, or used as a production admission threshold. We report sensitivity at the stricter thresholds 0.9 and 1.0. Unjudged candidates, missing summaries, and short-slate positions receive zero yield. Each metric is first computed per request event; duplicate events are then averaged within normalized request-text clusters, and clusters receive equal weight. We report 20,000 cluster-bootstrap replicates. The score comes from the same production LLM judge scorer used in the human alignment. Because the outcome counts score-qualified slots rather than human-labeled relevance among admissions, we call it *judge-qualified yield*, not human precision.

4.4 TTL-Aligned Candidate-Randomized Reranker Study

We analyze public-mode admitted impressions logged from July 17 through July 24, 2026. Candidates within a request were randomized between the standard candidate source (reranker off) and the direct-response LLM-reranker source (reranker on). The request is

therefore the analysis block. Content is deduplicated within each request and arm before computing outcomes.

Delivery-stack validation established that the reranker-on source has a three-day time-to-live (TTL), whereas the off source can continue dynamic retrieval through the seven-day Dear Algo lifetime. The primary comparison is therefore restricted to the common-support window $0 \leq \text{request-to-impression age} < 72$ hours. This restriction was adopted after operational validation and was not selected from the quality outcomes. The observable age begins at Dear Algo request authoring, while the serving TTL begins at message receipt, so the 72-hour window is a conservative proxy for the exact eligibility clock.

The window contains 93,972 logged rows across 4,321 requests. Of 1,531 paired-arm requests from 1,417 users, 1,346 requests from 1,278 users have at least one categorical LLM-as-a-judge label in both arms. After request-arm-item deduplication, the paired cohort contains 50,922 off candidates (31,099 judged) and 3,770 on candidates (2,928 judged); 845 requests from 823 users are fully judged in both arms.

The two sources retained their production candidate budgets. Because the agentic reranker-on path has higher serving computational cost and latency, it generated a substantially smaller candidate set per request; the 50,922-versus-3,770 imbalance is therefore expected by design rather than an allocation target or attrition. Candidate randomization occurred within request, but the paths did not operate over an equal-sized frozen slate. The study therefore compares end-to-end serving paths under their production candidate budgets.

For each request and arm, Exact-Relevant share is the fraction of judged admitted candidates labeled Relevant, and false-admission rate is the fraction labeled Irrelevant. Partial is neither a strict success nor a false admission. We first average candidates within request, then requests within user, and report the mean paired user contrast. Intervals use 20,000 user-cluster percentile-bootstrap replicates. Fully judged requests and same-request-session comparisons are sensitivities.

We also compute arbitrary-MNAR bounds over all 1,531 paired requests. For the lower Exact-Relevant bound, every unjudged on candidate is treated as not Relevant and every unjudged off candidate as Relevant; the upper bound reverses those assignments. False-admission bounds analogously assign every missing label adversarially. Each endpoint is reaggregated request-to-user and bootstrapped. Because rejected candidates and a frozen pre-reranking slate are unavailable, the study estimates admitted-set quality, not recall, NDCG, or rank movement.

4.5 Observational Explicit-to-Feed Study

We identify each user’s first eligible explicit-request day D after at least 28 days without another eligible request. We compare home-feed behavior in $D - 7, \dots, D - 1$ with $D + 1, \dots, D + 7$. The index day is excluded because the request timestamp is not trusted. We analyze three entry-point cohorts separately. In *public creation*, a user authors a public Threads post containing a Dear Algo request that other people can see. In *public repost*, a user reposts another person’s public Dear Algo request, adopting that request for the user’s own feed. In *private persisted intent*, a user adds a Dear Algo

Table 2: Human calibration of the strict categorical relevance gate ($N = 296$ evaluable audit pairs; $N_{\text{admitted}} = 89$).

Gate metric	Estimate	95% interval
Exact precision ^a	0.9438	[0.8876, 0.9888]
At-least-Partial precision ^b	1.0000	[0.9594, 1.0000]

Note: The primary metric is shown in bold.

^a The strict gate admits judge-Relevant items and counts human-Relevant items as successes: 84/89. Bootstrap interval.

^b The same 89 admissions are successful when human-Partial is also acceptable: 89/89. Exact Clopper–Pearson interval.

request privately, visible only to that user. These entry points differ in social visibility and request origination and may therefore select users with different motivations and baseline behavior. Separate reporting prevents cohort heterogeneity from being hidden by a pooled average, but it does not remove within-cohort self-selection.

The log has no native session identifier. We define a proxy session as a user/day/hour group and order its events by logged event time. The topline is the paired per-user change in proxy-session count. The negative-feedback endpoint is the number of report actions per 1,000 exposures, computed over the first 20 event-ordered exposures in each proxy session. Standard errors are clustered by index date.

All endpoints use two-sided 95% index-date-clustered intervals. Public creation and repost each have 54 index-date clusters; private persisted intent has 52. The intervals are descriptive and are not adjusted across request types or guardrails; no formal non-inferiority decision is assigned. A report-rate interval that includes zero is inconclusive, not evidence of non-inferiority: such a claim would require a prespecified maximum acceptable increase δ and an upper confidence bound below that margin.

5 Results

5.1 Human Calibration Supports a Narrow Strict Gate

Table 2 reports the strict-gate estimates.

Exact-Relevant precision was $84/89 = 0.9438$ (95% CI [0.8876, 0.9888]). The remaining five admissions were human-Partial and none was human-Irrelevant, yielding an at-least-Partial precision of $89/89 = 1.0000$ (exact 95% CI [0.9594, 1.0000]). In the stratified audit sample, the strict gate admitted 84 of 176 human-Relevant pairs; it assigned 76 to Partial and 16 to Irrelevant. This operating point trades coverage for admission precision, aligning with conservative persistent-feed admission rather than exhaustive retrieval.

Across the eligible frame, the locked judge labeled 32,526 of 48,876 records Relevant, corresponding to a 66.55% gate admission rate. Because the audit was stratified by judge label, its raw 89/296 admission fraction does not estimate this rate.

The post-hoc six-rater model-based sensitivity estimated exact precision at 0.8799 [0.7886, 0.9515], severe false admission at 0.0208 [0.0049, 0.1217], and at-least-Partial precision at 0.9792 [0.8783, 0.9951]. The seventh rater agreed with all 21 original majorities among 34 contested admissions and resolved 13 ties as nine Relevant and four Partial. An item-equal valid-vote check estimated exact precision at 0.8561 [0.7989, 0.9080] and Irrelevant share at 0.0379

[0.0114, 0.0712]. These sensitivities are reported beside, not instead of, the frozen labels.

The result is descriptive, not a preregistered pass. The precision-first estimand and threshold were clarified after label inspection. Initial human reliability was also weak: pairwise exact agreement was 0.677 and quadratic weighted kappa was 0.250 before the final-label rule.

5.2 The Full Configuration Yields More Judge-Qualified Candidates

Table 3 reports the paired configuration comparison.

At the 0.8 threshold, C reached 7.73 qualified candidates@20, adding 1.11 [0.12, 2.12] relative to B in the focal comparison and 2.15 [0.59, 3.73] relative to A in the contextual comparison. Across all three thresholds in Table 3, both the C–B and C–A intervals excluded zero. The C–B Yield@20 gains were +0.056, +0.056, and +0.049 at thresholds 0.8, 0.9, and 1.0, respectively; the corresponding C–A gains were +0.107, +0.097, and +0.079. Assigning zero yield to every arm for all eight malformed responses gives +0.0484 [0.0004, 0.0965]. The supported result is a configuration-level increase in fixed-denominator judge-qualified yield, not higher human precision or an independent grounding effect.

5.3 Lower False Admission Among TTL-Aligned Judged Admissions

Table 4 reports judged-admission quality within the common-support window.

Among judged admissions, the primary reranker-on minus off differences were -1.97 points [-3.02, -0.94] in Judge-Irrelevant share and +2.24 points [0.08, 4.41] in Exact-Relevant share. The fully judged sensitivity showed corresponding differences of -2.45 points [-3.92, -0.99] and +3.06 points [0.19, 5.86]. In the same request-session sensitivity, the differences were -0.27 points [-1.45, 0.91] and +2.69 points [-0.12, 5.53], respectively, with both intervals including zero. Within the shared 72-hour window, the reranker-on path improved both primary quality outcomes among judged admissions under production budgets.

5.4 Higher Later Usage with No Significant Change in Reports

Table 5 reports the primary all-eligible cohorts. Report rates are scaled per 1,000 eligible exposures. All intervals are two-sided and clustered by index date.

All three cohorts show higher post-period usage. Report-rate point estimates are lower in every cohort, but every two-sided interval includes zero. Excluding users with a repeat request retains the session pattern, while every report-rate interval again includes zero.

Together, these results show that explicit-request initiation is followed by higher Threads app usage, while report-rate changes are not statistically significant.

Table 3: Judge-qualified yield across score thresholds over $N = 72$ normalized request-text clusters with a fixed $K = 20$ slot budget.

Metric / score threshold	A: raw	B: derived	C: full	C-B [95% CI]	C-A [95% CI]
Yield@20, score ≥ 0.8	0.279	0.331	0.386	+0.056 [0.006, 0.106]	+0.107 [0.030, 0.186]
Yield@20, score ≥ 0.9	0.240	0.281	0.337	+0.056 [0.014, 0.099]	+0.097 [0.021, 0.175]
Yield@20, score ≥ 1.0	0.170	0.200	0.249	+0.049 [0.011, 0.088]	+0.079 [0.015, 0.146]
Qualified candidates@20, score ≥ 0.8	5.58	6.61	7.73	+1.11 [0.12, 2.12]	+2.15 [0.59, 3.73]

Note: The main reported score threshold is 0.8; 0.9 and 1.0 are stricter-threshold sensitivities. Yield@20 divides the number of qualified candidates by the fixed 20-slot denominator; the final row gives the equivalent count units at 0.8. C-B is the focal comparison and C-A is contextual. Boldface marks contrasts whose unadjusted 95% interval excludes zero. All intervals are cluster-bootstrap; all rows use 72 clusters.

Table 4: Candidate-randomized reranker quality within the delivery-eligible first 72 hours. Outcomes are evaluator-defined shares among judged admitted candidates.

Analysis	Outcome	Users / blocks	Reranker Off	Reranker On	Difference in points [95% CI]
Primary paired request	Judge-Irrelevant share	1,278 / 1,346	4.78%	2.80%	-1.97 [-3.02, -0.94]
Primary paired request	Exact-Relevant share	1,278 / 1,346	65.58%	67.82%	+2.24 [0.08, 4.41]
Fully judged request	Judge-Irrelevant share	823 / 845	5.65%	3.19%	-2.45 [-3.92, -0.99]
Fully judged request	Exact-Relevant share	823 / 845	63.46%	66.52%	+3.06 [0.19, 5.86]
Same request-session	Judge-Irrelevant share	949 / 1,306	3.12%	2.85%	-0.27 [-1.45, 0.91]
Same request-session	Exact-Relevant share	949 / 1,306	66.16%	68.86%	+2.69 [-0.12, 5.53]

Note: Primary pairs requests with at least one judged admission in each arm. Fully judged restricts to requests whose admitted candidates are all judged in both arms. Same request-session pairs the arms within a common request-session block. Users / blocks reports users and requests for the first two analyses, and users and request-session blocks for the last. Boldface marks contrasts whose 95% interval excludes zero. Intervals use 20,000 user-cluster bootstrap replicates. Lower Judge-Irrelevant share is better.

Table 5: Seven-day observational usage and report-rate changes after explicit-request initiation.

Entry point	Users	Pre sess.	Post sess.	Relative usage lift [95% CI]	Reports per 1,000 exposures			
					Pre	Post	Δ	Two-sided 95% CI
Public creation	40,376	37.26	39.63	+6.36% [5.67%, 7.04%]	0.036	0.027	-0.009	[-0.021, 0.003]
Public repost	47,412	35.98	37.47	+4.15% [3.30%, 4.99%]	0.018	0.017	-0.002	[-0.009, 0.005]
Private persisted intent	43,608	22.45	24.39	+8.61% [7.74%, 9.48%]	0.016	0.015	-0.001	[-0.007, 0.005]

Note: Sessions are average seven-day proxy-session counts per user around index day D . Reports are measured over the first 20 event-ordered exposures per session. Intervals are unadjusted two-sided 95% index-date-clustered intervals.

6 Discussion

6.1 Unification Is Layered, Not Binary

Dear Algo implements shared vocabulary, plan, and candidate interfaces for explicit and recommendation-derived intent. This architecture also illustrates why the two modes should not automatically share every operating point. An explicit request may tolerate a broad candidate stage followed by ranking; persistent recommendation intent may require a more conservative admission gate. A modular shared layer can limit negative transfer while still enabling cross-mode state.

6.2 Bridging IR and RecSys Evaluation

The offline candidate study resembles pooled IR evaluation: the request-time inventory and 20-slot budget are held fixed, while each configuration may retrieve a different slate that is compared

through paired resampling. The candidate-randomized study estimates a serving-path effect, but only on admitted, judge-labeled impressions inside arm eligibility; it cannot recover recall over the pre-reranking universe. The behavioral study resembles recommender-system analysis: it operates over users and time and measures sessions and report actions. These evaluations answer different questions. A unified system needs a contract that records request relevance, user outcomes, latency, computational cost, and safety rather than selecting whichever metric is favorable.

6.3 Future Work: Empirical Unification

The next step is to test whether carrying an explicit request into feed recommendation causally improves intent fulfillment. Eligible users or requests should be randomized between stored-plan enabled and disabled conditions while holding inventory, retrieval budgets, and downstream ranking fixed. The primary endpoint should be human intent fulfillment or a separately validated categorical evaluator on pooled, blinded candidates; app usage and report rate are secondary outcomes. Assignment, plan, inventory, fallback, and candidate-slate versions should be logged to support reproducible analysis.

7 Limitations and Responsible Use

The study has several principal limitations. First, the explicit-to-feed behavioral analysis is observational and uses hour-level proxy sessions; user self-selection, temporal trends, regression to the mean, and changes in feed composition may contribute to the observed differences. Second, all offline configurations in the paired candidate study use grounded tags and STQL, so that study cannot isolate their individual contributions. Third, human calibration covers public request-item pairs and does not establish private, per-intent, or multimodal validity; its post-hoc latent-label sensitivity is also prior- and rater-dependent. Fourth, the precision-first endpoints were clarified after initial result inspection, and the behavioral intervals are not multiplicity-adjusted. Fifth, the reranker study observes only admitted candidates: its TTL-aligned request-age window is a proxy for serving eligibility, judgment coverage and candidate composition differ by arm, arbitrary-MNAR bounds span both directions, and recall or pre-reranking rank movement cannot be estimated.

Persisted explicit or inferred intent may reveal sensitive interests or amplify an unwanted feedback loop. Production-system confidentiality and user-privacy constraints prevent release of implementation details and user-level logs; all reported results are aggregate, and we do not reproduce user requests. A deployable system requires data minimization, access control, expiration, deletion, and user correction. Negative preferences require collateral-suppression audits. Language, locale, visual-content, and low-history slices also require separate evaluation.

8 Conclusion

Dear Algo uses a shared, grounded, and executable intent layer for explicit and recommendation-like discovery while preserving task-specific serving choices. Human calibration supports a conservative categorical relevance gate, and an offline paired study finds higher judge-qualified candidate yield for the full retrieval configuration. Within its TTL-aligned delivery window, a candidate-randomized serving-path study finds a 1.97-percentage-point lower user-weighted judge-Irrelevant share and a 2.24-percentage-point higher Exact-Relevant share among judged admissions for the reranker on path under production budgets. Across three observational cohorts, explicit-request initiation was followed by 4.15%–8.61% higher subsequent Threads app usage as measured by proxy-session counts, while all report-rate confidence intervals included zero. Future randomized plan-on/off studies will estimate effects on intent fulfillment, with app usage and report rate as secondary outcomes.

The results support representation-level unification and precision-first evaluation, not yet causal cross-task transfer.

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References

- [1] Negar Arabzadeh and Charles L. A. Clarke. 2025. Benchmarking LLM-Based Relevance Judgment Methods. In *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '25)*. ACM, 3194–3204. <https://doi.org/10.1145/3726302.3730305>.
- [2] Negar Arabzadeh and Charles L. A. Clarke. 2025. A Human-AI Comparative Analysis of Prompt Sensitivity in LLM-Based Relevance Judgment. In *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '25)*. ACM, 2784–2788. <https://doi.org/10.1145/3726302.3730159>.
- [3] Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. 2024. Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection. In *The Twelfth International Conference on Learning Representations (ICLR '24)*. <https://openreview.net/forum?id=hSyW5go0v8>.
- [4] Krisztian Balog, Filip Radlinski, and Shushan Arakelyan. 2019. Transparent, Scrutable and Explainable User Models for Personalized Recommendation. In *Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '19)*. ACM, 265–274. <https://doi.org/10.1145/3331184.3331211>.
- [5] Anna Bavaresco, Raffaella Bernardi, Leonardo Bertolazzi, Desmond Elliott, Raquel Fernández, Albert Gatt, Esam Ghaleb, Mario Giulianelli, Michael Hanna, Alexander Koller, André Martins, Philipp Mondorf, Vera Neplenbroek, Sandro Pezzelle, Barbara Plank, David Schlangen, Alessandro Suglia, Aditya K. Surikuchi, Ece Takmaz, and Alberto Testoni. 2025. LLMs instead of Human Judges? A Large Scale Empirical Study across 20 NLP Evaluation Tasks. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers) (ACL '25)*. Association for Computational Linguistics, 238–255. <https://doi.org/10.18653/v1/2025.acl-short.20>.
- [6] Chris Buckley and Ellen M. Voorhees. 2004. Retrieval Evaluation with Incomplete Information. In *Proceedings of the 27th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '04)*. ACM, 25–32. <https://doi.org/10.1145/1008992.1009000>.
- [7] Micah Carroll, Adeline Foote, Kevin Feng, Marcus Williams, Anca Dragan, W. Bradley Knox, and Smitha Milli. 2025. CTRL-Rec: Controlling Recommender Systems With Natural Language. In *The Thirteenth International Conference on Learning Representations (ICLR '25)*. <https://openreview.net/forum?id=tXl1gdoAoV>.
- [8] Allison J. B. Chaney, Brandon M. Stewart, and Barbara E. Engelhardt. 2018. How Algorithmic Confounding in Recommendation Systems Increases Homogeneity and Decreases Utility. In *Proceedings of the 12th ACM Conference on Recommender Systems (RecSys '18)*. ACM, 224–232. <https://doi.org/10.1145/3240323.3240370>.
- [9] Li Chen and Pearl Pu. 2012. Critiquing-Based Recommenders: Survey and Emerging Trends. *User Modeling and User-Adapted Interaction* 22, 1–2, 125–150. <https://doi.org/10.1007/s11257-011-9108-6>.
- [10] Nicola De Cao, Gautier Izacard, Sebastian Riedel, and Fabio Petroni. 2021. Autoregressive Entity Retrieval. In *International Conference on Learning Representations (ICLR '21)*. <https://openreview.net/forum?id=5k8F6U39V>.
- [11] Laura Dietz, Oleg Zendel, Peter Bailey, Charles L. A. Clarke, Ellese Cotterill, Jeff Dalton, Faegheh Hasibi, Mark Sanderson, and Nick Craswell. 2025. Principles and Guidelines for the Use of LLM Judges. In *Proceedings of the 2025 International ACM SIGIR Conference on Innovative Concepts and Theories in Information Retrieval (ICTIR '25)*. ACM, 218–229. <https://doi.org/10.1145/3731120.3744588>.
- [12] Yann Dubois, Balázs Galambosi, Percy Liang, and Tatsunori B. Hashimoto. 2024. Length-Controlled AlpacaEval: A Simple Way to Debias Automatic Evaluators. *arXiv preprint arXiv:2404.04475*. <https://arxiv.org/abs/2404.04475>.
- [13] Miroslav Dudík, John Langford, and Lihong Li. 2011. Doubly Robust Policy Evaluation and Learning. In *Proceedings of the 28th International Conference on Machine Learning (ICML '11)*, 1097–1104. <https://arxiv.org/abs/1103.4601>.
- [14] Guglielmo Faggioli, Laura Dietz, Charles L. A. Clarke, Gianluca Demartini, Matthias Hagen, Claudia Hauff, Noriko Kando, Evangelos Kanoulas, Martin Potthast, Benno Stein, and Henning Wachsmuth. 2023. Perspectives on Large Language Models for Relevance Judgment. In *Proceedings of the 2023 ACM SIGIR International Conference on Theory of Information Retrieval (ICTIR '23)*. ACM, 39–50. <https://doi.org/10.1145/3578337.3605136>.
- [15] Maik Fröbe, Andrew Parry, Ferdinand Schlatt, Sean MacAvaney, Benno Stein, Martin Potthast, and Matthias Hagen. 2025. Large Language Model Relevance Assessors Agree With One Another More Than With Human Assessors. In *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '25)*. ACM, 2858–2863. <https://doi.org/10.1145/3726302.3730218>.
- [16] Luyu Gao, Xueguang Ma, Jimmy Lin, and Jamie Callan. 2023. Precise Zero-Shot Dense Retrieval without Relevance Labels. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers) (ACL '23)*. Association for Computational Linguistics. <https://doi.org/10.18653/>

- v1/2023.acl-long.99.
- [17] Siddharth Gollapudi, Neel Karia, Varun Sivashankar, Ravishankar Krishnaswamy, Nikit Begwani, Swapnil Raz, Yiyong Lin, Yin Zhang, Neelam Mahapatro, Premkumar Srinivasan, Amit Singh, and Harsha Vardhan Simhadri. 2023. Filtered-DiskANN: Graph Algorithms for Approximate Nearest Neighbor Search with Filters. In *Proceedings of the ACM Web Conference 2023 (WWW '23)*. ACM, 3406–3416. <https://doi.org/10.1145/3543507.3583552>.
- [18] Hui Huang, Xingyuan Bu, Hongli Zhou, Yingqi Qu, Jing Liu, Muyun Yang, Bing Xu, and Tiejun Zhao. 2025. An Empirical Study of LLM-as-a-Judge for LLM Evaluation: Fine-tuned Judge Model is not a General Substitute for GPT-4. In *Findings of the Association for Computational Linguistics: ACL 2025*. Association for Computational Linguistics, 5880–5895. <https://doi.org/10.18653/v1/2025.findings-acl.306>.
- [19] Xu Huang, Jianxun Lian, Yuxuan Lei, Jing Yao, Defu Lian, and Xing Xie. 2025. Recommender AI Agent: Integrating Large Language Models for Interactive Recommendations. *ACM Transactions on Information Systems* 43, 4, 1–33. <https://doi.org/10.1145/3731446>.
- [20] Dietmar Jannach, Ahtsham Manzoor, Wanling Cai, and Li Chen. 2021. A Survey on Conversational Recommender Systems. *ACM Computing Surveys* 54, 5, 1–36. <https://doi.org/10.1145/3453154>.
- [21] Thorsten Joachims, Adith Swaminathan, and Tobias Schnabel. 2017. Unbiased Learning-to-Rank with Biased Feedback. In *Proceedings of the Tenth ACM International Conference on Web Search and Data Mining (WSDM '17)*. ACM, 781–789. <https://doi.org/10.1145/3018661.3018699>.
- [22] Jaehun Jung, Faeze Brahman, and Yejin Choi. 2025. Trust or Escalate: LLM Judges with Provable Guarantees for Human Agreement. In *Proceedings of the 13th International Conference on Learning Representations (ICLR '25)*. <https://openreview.net/forum?id=UHPnqSTBPO>.
- [23] Seungone Kim, Jamin Shin, Yejin Cho, Joel Jang, Shayne Longpre, Hwaran Lee, Sangdo Yun, Seongjin Shin, Sungdong Kim, James Thorne, and Minjoon Seo. 2024. Prometheus: Inducing Fine-Grained Evaluation Capability in Language Models. In *Proceedings of the 12th International Conference on Learning Representations (ICLR '24)*. <https://openreview.net/forum?id=8euJaTveKw>.
- [24] Ron Kohavi, Roger Longbotham, Dan Sommerfield, and Randal M. Henne. 2009. Controlled Experiments on the Web: Survey and Practical Guide. *Data Mining and Knowledge Discovery* 18, 1, 140–181. <https://doi.org/10.1007/s10618-008-0114-1>.
- [25] Yehuda Koren. 2009. Collaborative Filtering with Temporal Dynamics. In *Proceedings of the 15th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '09)*. ACM, 447–456. <https://doi.org/10.1145/1557019.1557072>.
- [26] Walid Krichene and Steffen Rendle. 2020. On Sampled Metrics for Item Recommendation. In *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '20)*. ACM, 1748–1757. <https://doi.org/10.1145/3394486.3403226>.
- [27] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Kuttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. In *Advances in Neural Information Processing Systems 33 (NeurIPS '20)*. <https://proceedings.neurips.cc/paper/2020/hash/6b493220305f780e1bc26945df7481e5-Abstract.html>.
- [28] Lihong Li, Wei Chu, John Langford, and Xuanhui Wang. 2011. Unbiased Offline Evaluation of Contextual-Bandit-Based News Article Recommendation Algorithms. In *Proceedings of the Fourth ACM International Conference on Web Search and Data Mining (WSDM '11)*. ACM, 297–306. <https://doi.org/10.1145/1935826.1935878>.
- [29] Dongliang Liao, Chenxing Wang, and Yawen Zeng. 2026. Bridging Explicit and Implicit Intent: Unified Interest Generative Method for Joint Search-Recommendation Modeling. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*. ACM. <https://doi.org/10.1145/3774904.3792253>.
- [30] Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. 2023. G-Eval: NLG Evaluation using GPT-4 with Better Human Alignment. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP '23)*. Association for Computational Linguistics, 2511–2522. <https://doi.org/10.18653/v1/2023.emnlp-main.153>.
- [31] Wensheng Lu, Jianxun Lian, Wei Zhang, Guanghua Li, Mingyang Zhou, Hao Liao, and Xing Xie. 2024. Aligning Large Language Models for Controllable Recommendations. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers) (ACL '24)*. Association for Computational Linguistics, 8159–8172. <https://doi.org/10.18653/v1/2024.acl-long.443>.
- [32] Meta. 2026. Threads Dear Algo. *Meta Newsroom*. 11 February 2026. <https://about.fb.com/news/2026/02/threads-dear-algo/>. Accessed 26 July 2026.
- [33] Martin Mladenov, Elliot Creager, Omer Ben-Porat, Kevin Swersky, Richard Zemel, and Craig Boutilier. 2020. Optimizing Long-term Social Welfare in Recommender Systems: A Constrained Matching Approach. In *Proceedings of the 37th International Conference on Machine Learning (ICML '20)*. PMLR 119, 6987–6998. <https://proceedings.mlr.press/v119/mladenov20a.html>.
- [34] Sheshera Mysore, Mahmood Jasim, Andrew McCallum, and Hamed Zamani. 2023. Editable User Profiles for Controllable Text Recommendations. In *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '23)*. ACM, 993–1003. <https://doi.org/10.1145/3539618.3591677>.
- [35] Arjun Panickssery, Samuel R. Bowman, and Shi Feng. 2024. LLM Evaluators Recognize and Favor Their Own Generations. In *Advances in Neural Information Processing Systems 37 (NeurIPS '24)*. <https://arxiv.org/abs/2404.13076>.
- [36] Liana Patel, Peter Kraft, Carlos Guestrin, and Matei Zaharia. 2024. ACORN: Performant and Predicate-Agnostic Search Over Vector Embeddings and Structured Data. *Proceedings of the ACM on Management of Data* 2, 3, 1–27. <https://doi.org/10.1145/3654923>.
- [37] Gustavo Penha, Ali Vardasbi, Enrico Palumbo, Marco de Nadai, and Hugues Bouchard. 2024. Bridging Search and Recommendation in Generative Retrieval: Does One Task Help the Other? In *Proceedings of the 18th ACM Conference on Recommender Systems (RecSys '24)*. ACM. <https://doi.org/10.1145/3640457.3688123>.
- [38] Gustavo Penha, Edoardo D’Amico, Marco De Nadai, Enrico Palumbo, Alexandre Tamborrino, Ali Vardasbi, Max Lefarov, Shawn Lin, Timothy Heath, Francesco Fabbri, and Hugues Bouchard. 2025. Semantic IDs for Joint Generative Search and Recommendation. In *Proceedings of the Nineteenth ACM Conference on Recommender Systems (RecSys '25)*. ACM. <https://doi.org/10.1145/3705328.3759300>.
- [39] Shashank Rajput, Nikhil Mehta, Anima Singh, Raghunandan Hulikal Keshavan, Trung Vu, Lukasz Heldt, Lichan Hong, Yi Tay, Vinh Q. Tran, Jonah Samost, Maciej Kula, Ed H. Chi, and Maheswaran Sathiamoorthy. 2023. Recommender Systems with Generative Retrieval. In *Advances in Neural Information Processing Systems 36 (NeurIPS '23)*. https://proceedings.neurips.cc/paper_files/paper/2023/hash/20dcab0f14046a5c6b02b61da9f13229-Abstract-Conference.html.
- [40] Jerome Ramos, Hossein A. Rahmani, Xi Wang, Xiao Fu, and Aldo Lipani. 2024. Transparent and Scrutable Recommendations Using Natural Language User Profiles. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers) (ACL '24)*. Association for Computational Linguistics, 13971–13984. <https://doi.org/10.18653/v1/2024.acl-long.753>.
- [41] Torsten Scholak, Nathan Schucher, and Dzmitry Bahdanau. 2021. PICARD: Parsing Incrementally for Constrained Auto-Regressive Decoding from Language Models. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP '21)*. Association for Computational Linguistics, 9895–9901. <https://doi.org/10.18653/v1/2021.emnlp-main.779>.
- [42] Tobias Schnabel, Adith Swaminathan, Ashudeep Singh, Navin Chandak, and Thorsten Joachims. 2016. Recommendations as Treatments: Debiasing Learning and Evaluation. In *Proceedings of the 33rd International Conference on Machine Learning (ICML '16)*. PMLR 48, 1670–1679. <https://proceedings.mlr.press/v48/schnabel16.html>.
- [43] Guy Shani and Asela Gunawardana. 2011. Evaluating Recommendation Systems. In *Recommender Systems Handbook*. Springer, 257–297. https://doi.org/10.1007/978-0-387-85820-3_8.
- [44] Xiang Shen, Yuhang Zhou, Yifan Wu, Zhuokai Zhao, Siyu Lin, Lei Huang, Qianqian Zhong, Lizhu Zhang, Benyu Zhang, Xiangjun Fan, and Hong Yan. 2026. Agentic Recommender System with Hierarchical Belief-State Memory. *arXiv preprint arXiv:2605.14401*. <https://arxiv.org/abs/2605.14401>.
- [45] Teng Shi, Zihua Si, Jun Xu, Xiao Zhang, Xiaoxue Zang, Kai Zheng, Dewei Leng, Yanan Niu, and Yang Song. 2024. UniSAR: Modeling User Transition Behaviors between Search and Recommendation. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '24)*. ACM. <https://doi.org/10.1145/3626772.3657811>.
- [46] Teng Shi, Jun Xu, Xiao Zhang, Xiaoxue Zang, Kai Zheng, Yang Song, and Enyun Yu. 2025. GenSAR: Unifying Balanced Search and Recommendation with Generative Retrieval. In *Proceedings of the Nineteenth ACM Conference on Recommender Systems (RecSys '25)*. ACM. <https://doi.org/10.1145/3705328.3748071>.
- [47] Zihua Si, Zhongxiang Sun, Xiao Zhang, Jun Xu, Xiaoxue Zang, Yang Song, Kun Gai, and Ji-Rong Wen. 2023. When Search Meets Recommendation: Learning Disentangled Search Representation for Recommendation. In *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '23)*. ACM. <https://doi.org/10.1145/3539618.3591786>.
- [48] Ian Soboroff. 2025. Don’t Use LLMs to Make Relevance Judgments. *Information Retrieval Research* 1, 1. <https://doi.org/10.54195/irri.19625>.
- [49] Weiwei Sun, Lingyong Yan, Xinyu Ma, Shuaiqiang Wang, Pengjie Ren, Zhumin Chen, Dawei Yin, and Zhaochun Ren. 2023. Is ChatGPT Good at Search? Investigating Large Language Models as Re-Ranking Agents. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP '23)*. Association for Computational Linguistics. <https://doi.org/10.18653/v1/2023.emnlp-main.923>.
- [50] Zhongxiang Sun, Zihua Si, Xiaoxue Zang, Dewei Leng, Yanan Niu, Yang Song, Xiao Zhang, and Jun Xu. 2023. KuaiSAR: A Unified Search and Recommendation Dataset. In *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management (CIKM '23)*. ACM. <https://doi.org/10.1145/3583780.3615123>.
- [51] Jiakai Tang, Yujie Luo, Xunke Xi, Fei Sun, Xueyang Feng, Sunhao Dai, Chao Yi, Dian Chen, Zhujin Gao, Yang Li, Xu Chen, Wen Chen, Jian Wu, Yuning Jiang, and Bo Zheng. 2025. Interactive Recommendation Agent with Active User

- Commands. *arXiv preprint arXiv:2509.21317*. <https://arxiv.org/abs/2509.21317>.
- [52] Yi Tay, Vinh Q. Tran, Mostafa Dehghani, Jianmo Ni, Dara Bahri, Harsh Mehta, Zhen Qin, Kai Hui, Zhe Zhao, Jai Gupta, Tal Schuster, William W. Cohen, and Donald Metzler. 2022. Transformer Memory as a Differentiable Search Index. In *Advances in Neural Information Processing Systems 35 (NeurIPS '22)*. <https://openreview.net/forum?id=Vu-B0cIPfq>.
- [53] TechCrunch. 2026. Threads Adds New Personalization and Community Features as It Reaches 500M Monthly Users. 16 June 2026. <https://techcrunch.com/2026/06/16/threads-adds-new-personalization-and-community-features-as-it-reaches-500m-monthly-users/>. Accessed 26 July 2026.
- [54] The Verge. 2026. Threads' New "Dear Algo" Feature Lets You Tell the Algorithm What You Want to See. 11 February 2026. <https://www.theverge.com/tech/876610/meta-threads-dear-algo-algorithm>. Accessed 26 July 2026.
- [55] Paul Thomas, Seth Spielman, Nick Craswell, and Bhaskar Mitra. 2024. Large Language Models Can Accurately Predict Searcher Preferences. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '24)*. ACM, 1930–1940. <https://doi.org/10.1145/3626772.3657707>.
- [56] Harsh Trivedi, Niranjana Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2023. Interleaving Retrieval with Chain-of-Thought Reasoning for Knowledge-Intensive Multi-Step Questions. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers) (ACL '23)*. Association for Computational Linguistics. <https://doi.org/10.18653/v1/2023.acl-long.557>.
- [57] Ellen M. Voorhees. 2000. Variations in Relevance Judgments and the Measurement of Retrieval Effectiveness. *Information Processing & Management* 36, 5, 697–716. [https://doi.org/10.1016/S0306-4573\(00\)00010-8](https://doi.org/10.1016/S0306-4573(00)00010-8).
- [58] Bailin Wang, Richard Shin, Xiaodong Liu, Oleksandr Polozov, and Matthew Richardson. 2020. RAT-SQL: Relation-Aware Schema Encoding and Linking for Text-to-SQL Parsers. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL '20)*. Association for Computational Linguistics, 7567–7578. <https://doi.org/10.18653/v1/2020.acl-main.677>.
- [59] Liang Wang, Nan Yang, and Furu Wei. 2023. Query2doc: Query Expansion with Large Language Models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP '23)*. Association for Computational Linguistics. <https://doi.org/10.18653/v1/2023.emnlp-main.585>.
- [60] Peiyi Wang, Lei Li, Liang Chen, Zefan Cai, Dawei Zhu, Binghui Lin, Yunbo Cao, Lingpeng Kong, Qi Liu, Tianyu Liu, and Zhifang Sui. 2024. Large Language Models Are Not Fair Evaluators. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers) (ACL '24)*. Association for Computational Linguistics, 9440–9450. <https://doi.org/10.18653/v1/2024.acl-long.511>.
- [61] Xiaolei Wang, Kun Zhou, Ji-Rong Wen, and Wayne Xin Zhao. 2022. Towards Unified Conversational Recommender Systems via Knowledge-Enhanced Prompt Learning. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '22)*. ACM. <https://doi.org/10.1145/3534678.3539382>.
- [62] Yancheng Wang, Ziyan Jiang, Zheng Chen, Fan Yang, Yingxue Zhou, Eunah Cho, Xing Fan, Xiaojiang Huang, Yanbin Lu, and Yingzhen Yang. 2024. RecMind: Large Language Model Powered Agent for Recommendation. In *Findings of the Association for Computational Linguistics: NAACL 2024*. Association for Computational Linguistics, 4351–4364. <https://doi.org/10.18653/v1/2024.findings-naacl.271>.
- [63] Yueqi Wang, Yoni Halpern, Shuo Chang, Jingchen Feng, Elaine Ya Le, Longfei Li, Xujian Liang, Min-Cheng Huang, Shane Li, Alex Beutel, Yaping Zhang, and Shuchao Bi. 2023. Learning from Negative User Feedback and Measuring Responsiveness for Sequential Recommenders. In *Proceedings of the 17th ACM Conference on Recommender Systems (RecSys '23)*. ACM, 1049–1053. <https://doi.org/10.1145/3604915.3610244>.
- [64] Ledell Wu, Fabio Petroni, Martin Josifoski, Sebastian Riedel, and Luke Zettlemoyer. 2020. Scalable Zero-Shot Entity Linking with Dense Entity Retrieval. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP '20)*. Association for Computational Linguistics, 6397–6407. <https://doi.org/10.18653/v1/2020.emnlp-main.519>.
- [65] Jiayi Xie, Shang Liu, Gao Cong, and Zhenzhong Chen. 2024. UnifiedSSR: A Unified Framework of Sequential Search and Recommendation. In *Proceedings of the ACM Web Conference 2024 (WWW '24)*. ACM. <https://doi.org/10.1145/3589334.3645427>.
- [66] Bi Xue, Hong Wu, Lei Chen, Chao Yang, Yiming Ma, Fei Ding, Zhen Wang, Liang Wang, Xiaoheng Mao, Ke Huang, Xialu Li, Peng Xia, Rui Jian, Yanli Zhao, Yanzun Huang, Yijie Deng, Harry Tran, Ryan Chang, Min Yu, Eric Dong, Jiazhou Wang, Qianqian Zhang, Keke Zhai, Hongzhang Yin, Pawel Garbacki, Jiaqi Zhai, Zheng Fang, Yiyi Pan, Min Ni, Kevin Greer, Rui Zhang, and Yang Liu. 2026. SilverTorch: A Unified Model-based System to Democratize Large-Scale Recommendation on GPUs. In *Proceedings of the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '26)*. ACM. <https://doi.org/10.1145/3805712.3809755>.
- [67] Jing Yao, Zhicheng Dou, Ruobing Xie, Yanxiong Lu, Zhiping Wang, and Ji-Rong Wen. 2021. USER: A Unified Information Search and Recommendation Model based on Integrated Behavior Sequence. In *Proceedings of the 30th ACM International Conference on Information and Knowledge Management (CIKM '21)*. ACM. <https://doi.org/10.1145/3459637.3482489>.
- [68] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. ReAct: Synergizing Reasoning and Acting in Language Models. In *International Conference on Learning Representations (ICLR '23)*. https://openreview.net/forum?id=WE_vluYUL-X.
- [69] Hamed Zamani and W. Bruce Croft. 2018. Joint Modeling and Optimization of Search and Recommendation. In *Proceedings of the First Biennial Conference on Design of Experimental Search and Information Retrieval Systems (DESIRE '18)*. CEUR-WS.org. <https://ceur-ws.org/Vol-2167/paper2.pdf>.
- [70] Hamed Zamani and W. Bruce Croft. 2020. Learning a Joint Search and Recommendation Model from User-Item Interactions. In *Proceedings of the 13th International Conference on Web Search and Data Mining (WSDM '20)*. ACM. <https://doi.org/10.1145/3336191.3371818>.
- [71] Junjie Zhang, Ruobing Xie, Yupeng Hou, Wayne Xin Zhao, Leyu Lin, and Ji-Rong Wen. 2025. Recommendation as Instruction Following: A Large Language Model Empowered Recommendation Approach. *ACM Transactions on Information Systems* 43, 5, 1–37. <https://doi.org/10.1145/3708882>.
- [72] Yuting Zhang, Yiqing Wu, Ruidong Han, Ying Sun, Yongchun Zhu, Xiang Li, Wei Lin, Fuzhen Zhuang, Zhulin An, and Yongjun Xu. 2024. Unified Dual-Intent Translation for Joint Modeling of Search and Recommendation. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '24)*. ACM. <https://doi.org/10.1145/3637528.3671519>.
- [73] Kai Zhao, Yukun Zheng, Tao Zhuang, Xiang Li, and Xiaoyi Zeng. 2022. Joint Learning of E-commerce Search and Recommendation with a Unified Graph Neural Network. In *Proceedings of the Fifteenth ACM International Conference on Web Search and Data Mining (WSDM '22)*. ACM. <https://doi.org/10.1145/3488560.3498414>.
- [74] Yuyue Zhao, Jiancan Wu, Xiang Wang, Wei Tang, Dingxian Wang, and Maarten de Rijke. 2024. Let Me Do It For You: Towards LLM Empowered Recommendation via Tool Learning. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '24)*. ACM, 1796–1806. <https://doi.org/10.1145/3626772.3657828>.
- [75] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems 36 (NeurIPS '23)*. *Datasets and Benchmarks Track*. https://proceedings.neurips.cc/paper_files/paper/2023/hash/91f18a1287b398d378ef22505bf41832-Abstract-Datasets_and_Benchmarks.html.
- [76] Joyce Zhou, Weijie Zhou, Doug Turnbull, and Thorsten Joachims. 2026. SteerEval: A Framework for Evaluating Steerability with Natural Language Profiles for Recommendation. *arXiv preprint arXiv:2601.21105*. <https://arxiv.org/abs/2601.21105>.